

MODEL COST INTELLIGENCE

Architect Audit v5 | 2026-05-16 | Full Analysis

User Behavior Profile

AI Engineering Lead running a heavy multi-agent pipeline on flagship models. 60% coding, 25% research, 15% documentation. You run 15+ daily searches, compress every session, use skills 10x/day. Current stack: Opus 4.6 and Opus 4.7 for heavy tasks, GPT-4o and GPT-5.5 for extraction and reasoning, Gemini 3.1 Pro for triage. Target: same quality output at 90% lower cost.

Executive Summary & Strategic Roadmap

Unified model routing strategy based on 11 task modules.

TASK: DEFAULT CHAT

CURRENT: anthropic/claude-opus-4.6 (In: 5.00 / Out: 25.00)

OPTION A (Efficiency/Cost Leader): google/gemini-3.1-flash-lite | Input: \$0.25/1M | Output: \$1.50/1M | Context: 1M

OPTION B (Performance/Quality Leader): anthropic/claude-sonnet-4.6 | Input: \$3.00/1M | Output: \$15.00/1M | Context: 1M

FINAL RECOMMENDATION & DEEP RATIONALE:

Opus 4.6 at \$5/\$25 per 1M tokens is 20x the cost of Gemini 3.1 Flash Lite at \$0.25/\$1.50 for conversational chat. Flash Lite handles multi-turn dialogue, tool calls, and context with <350ms latency -- identical quality for daily chat. Claude Sonnet 4.6 (\$3/\$15) is still 12x more expensive than Flash Lite, offering no conversational quality advantage. Savings compound across 100+ daily interactions.

TASK: VISION ANALYSIS

CURRENT: openai/gpt-4o (In: 2.50 / Out: 10.00)

OPTION A (Efficiency/Cost Leader): google/gemini-3.1-flash-lite | Input: \$0.25/1M | Output: \$1.50/1M | Context: 1M

OPTION B (Performance/Quality Leader): openai/gpt-4o-mini | Input: \$0.15/1M | Output: \$0.60/1M | Context: 128K

FINAL RECOMMENDATION & DEEP RATIONALE:

GPT-4o at \$2.50/\$10 for Telegram screenshots and debugging images is heavily overprovisioned. Gemini 3.1 Flash Lite handles multimodal vision natively at \$0.25/\$1.50 with a massive 1M context window -- 10x cheaper. GPT-4o-mini (\$0.15/\$0.60) is even cheaper but lacks the 1M context that Flash Lite provides for large image batches. For debugging screenshots, Flash Lite is the clear winner.

TASK: WEB EXTRACT

CURRENT: openai/gpt-4o (In: 2.50 / Out: 10.00)

OPTION A (Efficiency/Cost Leader): google/gemini-3.1-flash-lite | Input: \$0.25/1M | Output: \$1.50/1M | Context: 1M

OPTION B (Performance/Quality Leader): anthropic/claude-sonnet-4.6 | Input: \$3.00/1M | Output: \$15.00/1M | Context: 1M

FINAL RECOMMENDATION & DEEP RATIONALE:

GPT-4o at \$2.50/\$10 for web parsing is unjustifiable when Flash Lite reads pages at \$0.25/\$1.50 with identical 1M context. With 10+ daily extractions, savings hit \$2.25-9.00 per day. Claude Sonnet 4.6 (\$3/\$15) is even more expensive, offering no extraction quality benefit over Flash Lite for structured content parsing.

TASK: COMPRESSION

CURRENT: anthropic/claude-opus-4.6 (In: 5.00 / Out: 25.00)

OPTION A (Efficiency/Cost Leader): inclusionai/ring-2.6-1t:free | Input: \$0.00/1M | Output: \$0.00/1M | Context: 128K

OPTION B (Performance/Quality Leader): google/gemini-3.1-flash-lite | Input: \$0.25/1M | Output: \$1.50/1M | Context: 1M

FINAL RECOMMENDATION & DEEP RATIONALE:

Opus 4.6 at \$5/\$25 for session compression is pure waste. Ring 2.6-1t:free is a zero-cost thinking model designed exactly for log summarization. No quality difference for compression tasks -- only cost savings from \$5/\$25 to \$0/\$0. Every compression run on Opus is silently burning budget.

TASK: SESSION SEARCH

CURRENT: openai/gpt-5.5 (In: 5.00 / Out: 30.00)

OPTION A (Efficiency/Cost Leader): meta-llama/llama-3.2-3b-instruct:free | Input: \$0.00/1M | Output: \$0.00/1M | Context: 128K

OPTION B (Performance/Quality Leader): google/gemini-3.1-flash-lite | Input: \$0.25/1M | Output: \$1.50/1M | Context: 1M

FINAL RECOMMENDATION & DEEP RATIONALE:

GPT-5.5 at \$5/\$30 for search retrieval is 2000x overpriced. Llama 3.2 3B Free handles 15+ daily queries at \$0.00 with sub-100ms latency. This is the single highest-impact switch -- pure cost elimination with zero quality loss. FTS5 keyword retrieval does not need frontier reasoning.

TASK: SKILLS HUB

CURRENT: anthropic/claude-opus-4.6 (In: 5.00 / Out: 25.00)

OPTION A (Efficiency/Cost Leader): meta-llama/llama-3.2-3b-instruct:free | Input: \$0.00/1M | Output: \$0.00/1M | Context: 128K

OPTION B (Performance/Quality Leader): mistralai/pixtral-12b | Input: \$0.15/1M | Output: \$0.15/1M | Context: 128K

FINAL RECOMMENDATION & DEEP RATIONALE:

10x/day skill matching on Opus 4.6 at \$5/\$25 burns budget silently. Free 3B handles text retrieval and skill matching instantly at \$0. No quality gap for simple keyword-to-skill mapping. Pixtral 12B (\$0.15/\$0.15) is a reasonable paid fallback but offers no advantage over free for this specific retrieval task.

TASK: APPROVAL

CURRENT: openai/gpt-5.5 (In: 5.00 / Out: 30.00)

OPTION A (Efficiency/Cost Leader): meta-llama/llama-3.2-3b-instruct:free | Input: \$0.00/1M | Output: \$0.00/1M | Context: 128K

OPTION B (Performance/Quality Leader): google/gemini-3.1-flash-lite | Input: \$0.25/1M | Output: \$1.50/1M | Context: 1M

FINAL RECOMMENDATION & DEEP RATIONALE:

Shell approval checks do not need frontier reasoning. Free 3B responds in sub-100ms vs GPT-5.5 at 800ms+. Safety logic for command approval is identical. Speed improvement reduces Telegram bot hang-time by 700ms per approval gate, and cost drops from \$5/\$30 to \$0/\$0.

TASK: MCP TOOLING

CURRENT: openai/gpt-4o (In: 2.50 / Out: 10.00)

OPTION A (Efficiency/Cost Leader): google/gemini-3.1-flash-lite | Input: \$0.25/1M | Output: \$1.50/1M | Context: 1M

OPTION B (Performance/Quality Leader): deepseek/deepseek-v3.1 | Input: \$0.15/1M | Output: \$0.75/1M | Context: 128K

FINAL RECOMMENDATION & DEEP RATIONALE:

GPT-4o at \$2.50/\$10 for tool trace analysis is 16x more expensive than DeepSeek V3.1 (\$0.15/\$0.75). DeepSeek V3.1 offers superior instruction-following for MCP tool parsing at a fraction of GPT-4o cost. Flash Lite (\$0.25/\$1.50) provides 1M vs 128K context advantage, but MCP traces rarely exceed 128K. Recommend DeepSeek V3.1 for low cost and strong agentic capabilities.

TASK: TITLE GENERATION

CURRENT: anthropic/claude-opus-4.6 (In: 5.00 / Out: 25.00)

OPTION A (Efficiency/Cost Leader): meta-llama/llama-3.2-3b-instruct:free | Input: \$0.00/1M | Output: \$0.00/1M | Context: 128K

OPTION B (Performance/Quality Leader): google/gemini-3.1-flash-lite | Input: \$0.25/1M | Output: \$1.50/1M | Context: 1M

FINAL RECOMMENDATION & DEEP RATIONALE:

Generating session titles with Opus 4.6 is a budget drain with zero ROI. Free 3B handles trivial naming at \$0 with identical title quality. Every dollar on Opus for this task is completely wasted -- title generation is a simple extraction task that even a 3B model handles perfectly.

TASK: TRIAGE SPECIFIER

CURRENT: google/gemini-3.1-pro-preview (In: 2.00 / Out: 12.00)

OPTION A (Efficiency/Cost Leader): google/gemini-3.1-flash-lite | Input: \$0.25/1M | Output: \$1.50/1M | Context: 1M

OPTION B (Performance/Quality Leader): anthropic/claude-sonnet-4.6 | Input: \$3.00/1M | Output: \$15.00/1M | Context: 1M

FINAL RECOMMENDATION & DEEP RATIONALE:

Gemini 3.1 Pro at \$2/\$12 is 8x more expensive than Flash Lite for repository routing. Context windows are identical at 1M. Flash Lite handles triage classification with 8x less cost and <350ms latency. Claude Sonnet 4.6 (\$3/\$15) is even more expensive -- both Pro and Sonnet are overkill for simple repo-to-task routing logic.

TASK: CURATOR

CURRENT: anthropic/claude-opus-4.7 (In: 5.00 / Out: 25.00)

OPTION A (Efficiency/Cost Leader): deepseek/deepseek-v3.1 | Input: \$0.15/1M | Output: \$0.75/1M | Context: 128K

OPTION B (Performance/Quality Leader): google/gemini-3.1-flash-lite | Input: \$0.25/1M | Output: \$1.50/1M | Context: 1M

FINAL RECOMMENDATION & DEEP RATIONALE:

Opus 4.7 at \$5/\$25 for memory pruning is excessive. DeepSeek V3.1 (\$0.15/\$0.75) provides superior instruction-following for curated deletion logic at 33x cheaper. Flash Lite (\$0.25/\$1.50) is also an option but DeepSeek V3.1 stronger reasoning at lower cost makes it the better fit for the curator decision-making workload.